CAD AI

Architectural & Mechanical 2D/3D Web CAD System — Complete Technical Specification, UI Layout, & Master Icon Catalog

Project Name:	cadai-app (CAD AI 2025)	**Core Engine:**	HTML5 Canvas + Konva.js + Three.js WebGL
Version:	1.0.0 Production Architecture	**Frontend Framework:**	React 19, TypeScript, Vite, Tailwind CSS
State Management:	Zustand (Canvas, Tool, Settings Stores)	**Export Formats:**	AutoCAD DXF (R2000), SVG, PNG, OBJ, .cadai
AI Assistant:	Claude Sonnet 4 Natural Language CAD Engine	**Document Type:**	Full System Specification & Icon Directory

Executive Summary:
CAD AI is an industrial-grade, browser-native Computer-Aided Design (CAD) application engineered for architects, civil engineers, interior planners, and mechanical designers. It bridges high-precision 2D drafting with instant 3D architectural model extrusion, smart AI generative drawing, layer isolation, snap-assisted geometry, and production-grade DXF/OBJ vector export. This document serves as the complete technical blueprint and exhaustive UI icon dictionary for the entire codebase.

Key System Highlights

- **Dual-Engine Pipeline:** Konva.js powers the high-performance 2D drafting canvas with sub-pixel snap and transformation handles, while Three.js WebGL renders realtime 3D architectural extrusions with orbit controls and 3D transform gizmos.
- **BIM Architectural Material Pipeline:** Vector 2D shapes are tagged with real-world 3D heights, base elevations, and material finishes (Concrete, Glass, Wood, Metal, Brick, CAD Gray) ready for 3D extrusion and Wavefront .OBJ export.
- **Full AutoCAD DXF Compatibility:** Generates standard AutoCAD DXF AC1015/R2000 text files compatible with AutoCAD, FreeCAD, LibreCAD, and SolidWorks.
- **Multi-Unit Civil Engineering Scaling:** Supports Metric (mm, cm, m, km) and Imperial (in, ft, yd) with scales from 1:1 up to 1:1000.
- **AI Generative Prompt Engine:** Converts natural language descriptions (e.g., 'Draw a 10x8 room with door and 2 windows') into exact CAD geometry.

1. System Architecture & Tech Stack

The application is architected around a reactive unidirectional data flow using Zustand stores, decoupled rendering canvases, and modular tool controllers.

Layer / Tier	Technologies	Key Responsibilities & Components
Presentation & UI	React 19, Tailwind CSS, Lucide React	TopMenuBar ribbon, LeftToolbar, PropertiesPanel, LayersPanel, StatusBar, interactive modal dialogs (Export, Templates, 3D Studio, Shortcuts).
2D Vector Engine	Konva.js 10.5, react-konva 19.2	Hardware-accelerated 2D stage rendering, selection marquee, live geometric snaps (endpoints, midpoints, centers), dimension lines, polyline vertex chaining, and transformation handles.
3D Extrusion Studio	Three.js 0.186, OrbitControls, TransformControls	WebGL 3D scene, dynamic 2D-to-3D polygon extrusion, procedural wall thickness, realistic PBR materials (glass refraction, concrete roughness, wood grain), 3D axis resize and translation gizmos.
State Management	Zustand 5.0 (3 dedicated stores)	• useCanvasStore: Layers, shapes, undo/redo history stacks, selection, camera zoom/pan, project meta. • useToolStore: Active drafting tool, default stroke/fill, stroke width, opacity, text styling. • useSettingsStore: Measurement units (Metric/Imperial), drawing scale ratios, distance formatting.
Geometry & Math	Custom TypeScript Vector Engine	screenToCanvas coordinate translation, distance calculation, bounding box computation, 3-point arc math, nearest snap point solver (endpoints, midpoints, centers, grid intersections).
Export & I/O Pipeline	AutoCAD DXF writer, SVG builder, Canvas PNG renderer, OBJ exporter	Serializes canvas state into standard AutoCAD DXF (AC1015), scalable vector graphics (SVG), high-res PNG bitmap, 3D Wavefront (.OBJ), and native .cadai JSON project files.
AI Generative Service	Anthropic Claude API + Offline Rule Fallback	Translates natural language text prompts into structured CAD primitives with coordinate offset calculation and immediate layer insertion.

2. Project Directory & File Map

The codebase adheres to strict modular separation of concerns:

Directory / File	Role & Core Responsibility
`src/pages/EditorPage.tsx`	Main CAD workspace assembly uniting header, canvas, side panels, status bar, and autosave restore.
`src/components/toolbar/TopMenuBar.tsx`	Ribbon menu system featuring 7 functional tabs (Home, Draw, Modify, Align, View, Layers, Output) and Title Bar.
`src/components/toolbar/LeftToolbar.tsx`	Vertical drafting tools dock (Select, Move, Eraser, Pan, Line, Rect, Circle, Ellipse, Arc, Polyline, Text, Dimension).
`src/components/canvas/Canvas.tsx`	Konva 2D stage with event listeners for draw, pan (including Space-pan), select, drag, snap detection, and delete.
`src/components/panels/PropertiesPanel.tsx`	Sidebar for numeric coordinates, dimensions, stroke/fill styling, and 3D extrusion/material parameters.
`src/components/panels/LayersPanel.tsx`	CAD layer manager with visibility toggle, lock/unlock, custom color assignment, and layer renaming.
`src/components/panels/AIPanel.tsx`	Floating AI prompt dialog interfacing with Claude Sonnet and offline geometry interpreter.
`src/components/ui/StatusBar.tsx`	Bottom bar hosting AutoCAD command input bar, live X/Y coordinates, shape counts, and GRID/SNAP/Unit toggles.
`src/components/ui/ThreeDViewerModal.tsx`	Three.js WebGL 3D Extrusion studio with orbit controls, 3D gizmos, dimension adjustments, and OBJ export.
`src/components/ui/TemplatesModal.tsx`	Architectural floor plan templates modal with instant 2D draft load or 3D launch.

`src/components/ui/ExportModal.tsx`	Save & Export dialog for .cadai JSON, DXF, SVG, and PNG.
`src/components/ui/KeyboardShortcutsModal.tsx`	Searchable hotkey directory dialog covering tools, edit, view, file, and alignment.
`src/utils/exportUtils.ts`	Mathematical serializing algorithms for DXF (AC1015 format), SVG XML, and PNG rendering.
`src/utils/geometry.ts`	Computational geometry algorithms: bounding boxes, snap point detection, 3-point arcs, distances.

3. Master Icon Directory & UI Placement Guide

This exhaustive catalog details every icon present in the project, its visual component representation, exact screen placement/position, function performed, triggered store action, and keyboard shortcut.

3.1 Top Title Bar (Quick Access & Global Controls)

UI Position	Icon / Visual	Function & Action Performed	Store / Target Action	Shortcut
Top Header, far left	`Badge 'C'`	Application brand identity badge for CAD AI.	Visual Branding	—
Top Header, left cluster	`Save (Save)`	Opens the Save & Export Center modal to save .cadai or export files.	setShowExportModal(true)	Ctrl+S
Top Header, left cluster	`FolderOpen`	Launches native file picker to open and load an existing .cadai project JSON.	loadProjectFromFile()	Ctrl+O
Top Header, left cluster	`Undo2`	Rolls back canvas state to previous history snapshot. Disabled when history empty.	useCanvasStore.undo()	Ctrl+Z
Top Header, left cluster	`Redo2`	Re-applies most recently undone state from future stack. Disabled when future empty.	useCanvasStore.redo()	Ctrl+Y
Top Header, center title	`Pencil`	Reveals inline editable text input to rename the active project across the app.	setProjectName(name)	Enter
Top Header, center	`Dot (Amber/Green)`	Live autosave state: Pulsing amber = 'Saving...', Glowing green = 'Saved' in localStorage.	saveStatus indicator	—
Top Header, right zoom	`Minus '−'`	Decrements canvas zoom level by 10% (zoom - 0.1).	setZoom(zoom - 0.1)	-
Top Header, right zoom	`Plus '+'`	Increments canvas zoom level by 10% (zoom + 0.1).	setZoom(zoom + 0.1)	+
Top Header, right zoom	`Button '1:1'`	Resets viewport zoom to exactly 100% (1:1 pixel scale ratio).	setZoom(1.0)	0
Top Header, right zoom	`Button 'Fit'`	Calculates bounding box of all shapes across all visible layers and fits viewport.	zoomToFit(false)	F
Top Header, right side	`Sparkles`	Opens Starter Architectural Templates Modal (emerald themed button).	setShowTemplatesModal(true)	—
Top Header, right side	`Box (3D)`	Prominent cyan-blue glowing button: Launches the 3D WebGL Extrusion Studio.	setShow3DModal(true)	—
Top Header, far right	`HelpCircle`	Opens the comprehensive Keyboard Shortcuts Reference Modal.	setShowShortcutsModal(true)	?

3.2 Left Vertical Toolbar (Primary Drafting Tools)

UI Position	Icon / Visual	Function & Action Performed	Store / Target Action	Shortcut
Left Bar, Item 1	`MousePointer2`	Activates Select Tool. Drag bounding box to multi-select; click to select; drag handles to resize.	setActiveTool('select')	V
Left Bar, Item 2	`Move`	Activates Move Tool. Drag any selected shape or shape group across the canvas.	setActiveTool('move')	M
Left Bar, Item 3	`Eraser`	Activates Eraser Tool. Click directly on any shape or dimension line on canvas to delete it.	setActiveTool('eraser')	X
Left Bar, Item 4	`ZoomIn (Pan)`	Activates Pan Tool. Left-click and drag anywhere on canvas to pan. (Or hold Space key).	setActiveTool('pan')	H
Left Bar, Item 5	`Minus (Line)`	Activates Line Tool. Click start point and end point (with geometric snap) to draw line.	setActiveTool('line')	L
Left Bar, Item 6	`Square (Rect)`	Activates Rectangle Tool. Click and drag diagonally to draw 4-corner rectangle.	setActiveTool('rectangle')	R
Left Bar, Item 7	`Circle`	Activates Circle Tool. Click center point and drag to set radius with live snap feedback.	setActiveTool('circle')	C
Left Bar, Item 8	`PenTool (Ellipse)`	Activates Ellipse Tool. Click and drag to draw ellipse with width (rx) and height (ry).	setActiveTool('ellipse')	E
Left Bar, Item 9	`ArcIcon (SVG Arc)`	Activates 3-Point Arc Tool. Click center point, radius start point, and sweep end angle.	setActiveTool('arc')	A
Left Bar, Item 10	`Slash (Polyline)`	Activates Polyline Tool. Click consecutive points to chain line segments. Press Enter to end.	setActiveTool('polyline')	P
Left Bar, Item 11	`Type (Text)`	Activates Text Tool. Click canvas to place text annotation with inline font editor.	setActiveTool('text')	T
Left Bar, Item 12	`Ruler (Dimension)`	Opens Dimension Settings flyout (Units & Scale) with badge of active unit (e.g. mm).	showDimPanel / setActiveTool('dimension')	D

3.3 Top Ribbon Tabs (Action Command Ribbon)

UI Position	Icon / Visual	Function & Action Performed	Store / Target Action	Shortcut
Ribbon: Modify Group	`Copy`	Duplicates selected shapes with an automatic 20mm visual offset.	copyShapes()	**Ctrl+C / D**
Ribbon: Modify Group	`RotateCw (Rot ■)`	Rotates selected shapes 90 degrees clockwise around their geometric center.	rotateSelectedShapes(90)	**Ctrl+→**
Ribbon: Modify Group	`RotateCcw (Rot ■)`	Rotates selected shapes 90 degrees counter-clockwise around center.	rotateSelectedShapes(-90)	**Ctrl+←**
Ribbon: Modify Group	`FlipHorizontal (Mir H)`	Mirrors selected shapes horizontally across their central vertical axis.	mirrorSelectedShapes('h')	—
Ribbon: Modify Group	`MirrorVIcon (Mir V)`	Mirrors selected shapes vertically across their central horizontal axis.	mirrorSelectedShapes('v')	—
Ribbon: Modify Group	`Trash2 (Delete)`	Deletes all currently selected shapes or dimension annotations from canvas.	deleteSelectedShapes()	**Del / Backspace**
Ribbon: View Group	`Grid3x3 (Grid)`	Toggles millimeter background drafting grid visibility on/off.	toggleGrid()	**G**
Ribbon: View Group	`Magnet (Snap)`	Toggles smart snapping (grid intersections, line endpoints, midpoints, shape centers).	toggleSnap()	**S**
Ribbon: View Group	`Maximize2 (Fit All)`	Adjusts zoom and pan so all shapes on all visible layers fit into screen.	zoomToFit(false)	**F**
Ribbon: View Group	`Focus (Fit Sel)`	Adjusts zoom and pan to zoom specifically into currently selected shapes.	zoomToFit(true)	**Shift+F**
Ribbon: Align Group	`AlignLeftIcon (Left)`	Aligns left bounding edges of 2+ selected shapes to the leftmost shape.	alignSelectedShapes('left')	—
Ribbon: Align Group	`AlignCenterHIcon (Cntr H)`	Aligns horizontal center X coordinates of 2+ selected shapes.	alignSelectedShapes('centerH')	—
Ribbon: Align Group	`AlignRightIcon (Right)`	Aligns right bounding edges of 2+ selected shapes to the rightmost shape.	alignSelectedShapes('right')	—
Ribbon: Align Group	`AlignTopIcon (Top)`	Aligns top bounding edges of 2+ selected shapes to the topmost shape.	alignSelectedShapes('top')	—
Ribbon: Align Group	`AlignCenterVIcon (Cntr V)`	Aligns vertical center Y coordinates of 2+ selected shapes.	alignSelectedShapes('centerV')	—
Ribbon: Align Group	`AlignBottomIcon (Bottom)`	Aligns bottom bounding edges of 2+ selected shapes to the bottommost shape.	alignSelectedShapes('bottom')	—
Ribbon: Align Group	`DistributeHIcon (Dist H)`	Distributes 3+ selected shapes evenly along the horizontal X axis with equal gaps.	distributeSelectedShapes('h')	—
Ribbon: Align Group	`DistributeVIcon (Dist V)`	Distributes 3+ selected shapes evenly along the vertical Y axis with equal gaps.	distributeSelectedShapes('v')	—
Ribbon: Align Group	`GroupIcon (Group)`	Combines 2+ selected shapes into an atomic group with a shared groupId.	groupSelectedShapes()	**Ctrl+G**
Ribbon: Align Group	`UngroupIcon (Ungroup)`	Splits selected group back into individual autonomous shapes.	ungroupSelectedShapes()	**Ctrl+Shift+G**
Ribbon: Align Group	`Lock / Unlock`	Locks or unlocks selected shapes to prevent accidental moving or editing.	toggleLockSelectedShapes()	**Ctrl+Shift+L**
Ribbon: Output Group	`FileCode (DXF)`	Generates and downloads AutoCAD DXF AC1015/R2000 file from all layers.	exportAsDXF()	—
Ribbon: Output Group	`Download (SVG)`	Exports scalable vector graphics (SVG) with layers and color preservation.	exportAsSVG()	—
Ribbon: Output Group	`Image (PNG)`	Renders canvas to high-resolution raster PNG image file.	exportAsPNG()	—

3.4 Properties & Layers Panels (Inspector & Layer Stack)

UI Position	Icon / Visual	Function & Action Performed	Store / Target Action	Shortcut
Properties Panel, Top	`Tabs: 'properties' / 'layers'`	Switches between geometric property inspector and multi-layer CAD manager.	setActiveTab('properties'|'layers')	—
Properties, 3D section	`Box (3D Extrusion)`	Inspects 3D Height (Z), Elevation (Base Z), and opens 3D Studio directly.	updateShape(id, { height3d, elevation3d })	—
Properties, Materials	`■ Concrete / ■ Glass`	Applies architectural PBR material finish to the selected object for 3D extrusion.	updateShape(id, { material3d: 'concrete'|'glass' })	—
Properties, Materials	`■ Wood / ■■ Metal / ■ Brick`	Assigns realistic Wood, Metal, or Brick material shaders in 3D WebGL viewer.	updateShape(id, { material3d: 'wood'|'metal'|'brick' })	—
Properties, Bottom	`Trash2 (Erase / Delete)`	Deletes the currently inspected shape immediately.	deleteShape(selectedShape.id)	**Del**
Layers Panel, Header	`Plus (New Layer)`	Creates a new drafting layer with a distinctive auto-assigned CAD color.	addLayer()	—
Layers Panel, Row hover	`Eye / EyeOff`	Toggles layer visibility on 2D canvas and 3D extrusion scene.	toggleLayerVisibility(layer.id)	—
Layers Panel, Row hover	`Lock / Unlock`	Locks or unlocks the layer to protect all child shapes from alteration.	toggleLock(layer.id)	—
Layers Panel, Row hover	`Trash2 (Delete Layer)`	Deletes layer and all shapes inside it (disabled if only 1 layer exists).	deleteLayer(layer.id)	—

3.5 Status Bar & AI Draw Floating Widget

UI Position	Icon / Visual	Function & Action Performed	Store / Target Action	Shortcut
Status Bar, Command row	`Input: 'Command:'`	AutoCAD-style command interpreter: type L, R, C, E, P, D, T, M, H, GRID, SNAP.	executeCommand(cmd)	**Enter / Space**
Status Bar, Bottom right	`Text 'GRID'`	Toggles grid display; turns bright green when active.	toggleGrid()	**G**
Status Bar, Bottom right	`Text 'SNAP'`	Toggles snap engine; turns bright green when active.	toggleSnap()	**S**
Status Bar, Bottom right	`Settings (Unit/Scale)`	Displays active unit (e.g. mm) and scale (1:1); opens Unit & Scale Modal.	UnitSelector / setUnit() / setScale()	—
Canvas Overlay, bottom-right	`Sparkles ('AI Draw')`	Opens the AI Prompt Drawer to describe drawings in natural language.	setIsOpen(true)	—
AI Draw Panel, footer	`Send / Loader2`	Sends prompt to Claude API or offline fallback to generate CAD vector shapes.	handleGenerate() -> addShape()	**Enter**
AI Draw Panel, header	`Close 'X'`	Collapses AI Draw dialog back into floating button.	setIsOpen(false)	**Esc**

3.6 3D Architectural Extrusion Studio Modal

UI Position	Icon / Visual	Function & Action Performed	Store / Target Action	Shortcut
3D Modal, Header left	`Box (3D Cube)`	Studio header displaying live 3D mesh count.	ThreeDViewerModal Header	—
3D Modal, Header right	`Plus (+ Add 3D Block)`	Creates a new volumetric architectural 3D block directly in the scene.	handleAdd3DBlock()	—
3D Modal, Header right	`Download (Export .OBJ)`	Generates and downloads standard Wavefront .OBJ 3D file for Blender/AutoCAD.	handleExportOBJ()	—
3D Modal, Header right	`Maximize2 / Minimize2`	Toggles true full-screen mode for immersive 3D architectural review.	toggleFullscreen()	F
3D Modal, Header right	`Close 'X'`	Exits 3D Extrusion studio and returns to 2D drafting canvas.	setShow(false)	Esc
3D View, Top-Left Bar	`Ruler (Resize Dimensions)`	Activates interactive 3D axis resize handles directly on 3D blocks (Red X, Blue Y, Green Z).	setGizmoMode('scale')	S
3D View, Top-Left Bar	`Move (3D Move Arrows)`	Activates 3D translation arrows to move blocks in X, Y, or Z elevation.	setGizmoMode('translate')	M / G
3D Inspector, Navigation	`ChevronUp / Down / Left / Right`	4-Way directional D-Pad: Moves block Forward, Back, Left, or Right by 25mm steps.	moveShapeBy(dx, dy)	—
3D Inspector, Bottom	`Trash2 (Erase from 3D)`	Deletes inspected 3D block from both 3D scene and 2D canvas.	deleteShape(id)	Del
3D Controls, Bottom bar	`Compass (View Presets)`	Switches camera between Isometric, Top Plan, Front Elevation, and Side Elevation.	setCameraView('iso'|'top'|'front'|'side')	—
3D Controls, Bottom bar	`RotateCcw (Reset Camera)`	Resets 3D camera position, orbit angle, and target to default perspective.	handleResetCamera()	—

3.7 Dialogs & Modals (Templates, Export, Shortcuts, Units)

UI Position	Icon / Visual	Function & Action Performed	Store / Target Action	Shortcut
Templates Modal, Card action	`Box (Load & Launch 3D)`	Loads architectural floor plan template and immediately opens in 3D Extrusion studio.	handleLoad(template, true)	—
Templates Modal, Card action	`ArrowRight / Check`	Loads floor plan template directly onto 2D canvas with automatic zoom-to-fit.	handleLoad(template, false)	—
Export Modal, Save row	`FileJson (Save Project)`	Serializes full canvas, layers, and settings into a downloadable .cadai JSON file.	saveProjectToFile()	Ctrl+S
Export Modal, Export rows	`FileCode / Download / Image`	Exports active drawing to AutoCAD DXF, Scalable Vector Graphics (SVG), or PNG Bitmap.	exportAsDXF() / SVG / PNG	—
Shortcuts Modal, Search	`Search`	Instant search input filtering keyboard shortcuts by action name or key combination.	setSearchQuery(query)	—
Unit Selector Modal, Grids	`Check`	Visual checkmark highlighting the currently active unit (e.g. mm) or scale (e.g. 1:1).	setUnit(u) / setScale(s)	—

4. Key Features & Recent Architectural Fixes

This section highlights the 3 critical user-requested fixes and enhancements implemented in the codebase:

Issue 1: Deletion of Measurements and Dimensions

- **Root Cause:** When users placed dimension annotations on canvas, pressing Delete or Backspace did not trigger deletion because the global keydown listener was not wired to deleteSelectedShapes().
- **Resolution Implemented:** Added Delete and Backspace keyboard bindings in Canvas.tsx and useKeyboard.ts. Dimensions can now be selected with the Select tool (V) and deleted instantly with Delete, Backspace, the Ribbon Delete button, or the Right Properties panel 'Erase / Delete Shape' button.
- **Eraser Tool:** The dedicated Eraser Tool (X) on the Left Toolbar also allows clicking directly on any dimension or shape to eliminate it in one click.

Issue 2: Canvas Panning and Movement System

- **Root Cause:** Middle-mouse button auto-cancel guard interfered with the Pan tool, and the industry-standard Space-bar panning shortcut was missing.
- **Resolution Implemented:**
  1. **Space-Key Pan (Industry Standard):** Hold the Space bar from any active tool to temporarily transform the cursor into a 'grab' hand and drag anywhere to pan smoothly.
  2. **Pan Tool (H):** Left-click and drag anywhere on the canvas.
  3. **Middle Mouse Drag:** Click and drag with mouse wheel button.
  4. **Mouse Wheel Zoom:** Zoom in/out centered at cursor position.
  5. **Zoom to Fit (F):** Auto-centers all shapes instantly.

Issue 3: 3D View Studio Button Visibility & Placement

- **Root Cause:** The 3D View button was located in an unobtrusive corner where users frequently missed it.
- **Resolution Implemented:**
  1. Relocated to the **Top Menu Bar Title Bar (Right Section)**, right next to the Templates button.
  2. Styled with a striking **cyan-to-blue gradient** (from-cyan-600 to-blue-600) with glowing drop-shadow (shadow-[0_0_8px_rgba(6,182,212,0.45)]).
  3. Added hover glow intensification, Box 3D icon, and smooth active scale animations.
  4. Redundant access points preserved in Ribbon 'View' tab, Ribbon 'Output' tab, and Properties Panel '3D Extrusion' inspector.

5. Complete Keyboard Shortcuts Matrix

Category	Key Combination	Action Description
Drafting Tools	**V**	Select tool (Marquee drag, resize, rotate)
Drafting Tools	**M**	Move tool (Drag selected shapes)
Drafting Tools	**X**	Eraser tool (Click any shape/dimension to erase)
Drafting Tools	**H / Space + Drag**	Pan canvas (Drag viewport smoothly)
Drafting Tools	**L / R / C / E**	Line / Rectangle / Circle / Ellipse drafting
Drafting Tools	**A / P / T / D**	3-Point Arc / Polyline / Text / Dimension tool
Editing & History	**Ctrl + Z / Ctrl + Y**	Undo / Redo canvas actions
Editing & History	**Ctrl + C / Ctrl + D**	Copy / Duplicate selected shapes
Editing & History	**Delete / Backspace**	Delete selected shapes or dimensions
Editing & History	**Ctrl + G / Ctrl+Shift+G**	Group / Ungroup selected shapes
Editing & History	**Ctrl + Shift + L**	Lock / Unlock selected shapes
Editing & History	**Ctrl + → / Ctrl + ←**	Rotate 90° Clockwise / Counter-Clockwise
Navigation & View	**F / Shift + F**	Zoom to Fit All / Zoom to Fit Selection
Navigation & View	**+ / - / 0**	Zoom In / Zoom Out / Reset Zoom to 100%
Navigation & View	**G / S**	Toggle Grid / Toggle Snap
File & Modals	**Ctrl + S / Ctrl + O**	Save & Export / Open .cadai project
File & Modals	**? / Esc**	Keyboard Shortcuts Help / Cancel or Deselect